WOX7001 RESEARCH METHODOLOGY 1/2025/2026

Temporal and Weather Feature Integration for Spatio-Temporal Graph Neural Networks Metro Ridership Prediction in Malaysia's Tropical Climate

RESEARCH PROPOSAL

Matric Number:	24087980
Name	Faiz Yah Bin Abdul Hanan Yah
Occurrence:	7
Lecturer Name:	Assoc. Prof. Dr. Norjihan Abdul Ghani

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Abstract

Urban rail transit systems are expanding rapidly in tropical Southeast Asian cities which calls for accurate ridership prediction models for effective operational planning and resource allocation. Spatio-Temporal Graph Neural Networks (STGNNs) have demonstrated promising performance for metro ridership prediction, yet existing research exhibits substantial geographical concentration in temperate climate regions (China, USA, Netherlands) where weather shows higher variability. The generalisability of temporal and weather features to tropical contexts remains empirically unvalidated, while the literature lacks systematic ablation studies isolating individual feature contributions and feature importance analysis. This research addresses these gaps through a three-variant ablation study evaluating STGNN performance in Malaysia's tropical climate, baseline model using only historical ridership and spatial structure (M1), baseline with temporal features (M2), and baseline with temporal and weather features (M3). The architecture of STGNN is a combination of GCN spatial architecture with GRU temporal architecture, utilising a 23-hour lookback window to predict next-hour, modelled on KTM Komuter data 2024. The study quantifies incremental contributions of each feature category through standard regression metrics (MAE, RMSE, MAPE, R^2) and identifies strongest predictors via SHAP and permutation importance methods. The methodology is tested on KTM Komuter annual ridership data from 2024, encompassing station-level transactions, meteorological variables, and Malaysian holiday calendars. The findings are expected to provide empirical validation of temperate-climate assumptions in tropical settings, quantified guidance on optimal feature selection for computational efficiency, and climate-specific importance rankings. This research contributes to understanding STGNN generalisability across diverse geographical contexts while providing actionable insights for transit operators on cost-effective model deployment, supporting SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action) through improved prediction accuracy and optimised operations.

Keywords: Spatio-Temporal Graph Neural Networks, Metro Ridership Prediction, Tropical Climate, Feature Importance Analysis, Ablation Study

1 Introduction

1.1 Research Background

Urban rail transit systems play a critical role in ensuring mobility of the population, especially in densely populated metropolitan areas. An accurate metro ridership prediction is vital for optimisation of resource allocation, improvement of service quality and enhancement of operational efficiency. Recent advances in deep learning, particularly Graph Neural Networks (GNNs) have shown promising performance in capturing spatio-temporal dependencies inherent in metro ridership (Jin et al. 2024; Li et al. 2025; Wang et al. 2024), since the connections between stations represent the spatial pattern while the hourly ridership reflects the temporal patterns.

Spatio-Temporal Graph Neural Networks (STGNNs) have emerged as a strong architecture for transit ridership prediction with its capacity in capturing the relationship of inherent spatial nature of metro connections combined with the temporal dynamics of the hourly ridership. Short-term ridership prediction for metro systems typically uses historical data from 12 to 24 hours to forecast demand 10 to 60 minutes ahead, enabling effective operational planning and real-time resource allocation (Jin et al. 2024). Studies such as Multi-dimensional Time-dependent Dynamic Graph Neural Network, (MTDGNN) (Li et al. 2025), Dynamic Spatio-Temporal Graph Fusion Network (DSTGFN) (Liu et al. 2025) have demonstrated improved performance compared to traditional time series methods. These models leverage graph structure of metro networks to learn spatial dependencies and integrate temporal modules to learn the ridership patterns across time.

However, there exists several gaps in the current body of literature. Firstly, a large majority of STGNN research has been using datasets from regions with temperate climate, predominantly in China and the United States. Cities in these countries such as Beijing, Shanghai and New York have significant temperature changes across the year which has an impact on the ridership patterns. Secondly, while some studies have incorporated external features such as weather and temporal context (Farahmand, Gkiotsalitis, and Geurs 2023), there remain gaps in systematic analysis on how features contribute to prediction performance. Thirdly, even if ablation study was implemented, there is limited empirical evidence on whether similar features that contribute in temperate climates remain relevant in tropical contexts.

As the majority of research was conducted with temperate climate dataset, Malaysia poses an unique opportunity for validating the STGNN model in context with tropical climate characteristics. Unlike temperate regions with drastic temperature changes, Malaysia has a relatively stable temperature throughout the year, high humidity and distinct rainfall patterns. It is crucial to then identify if weather-based features that improve prediction in temperate climates (Farahmand et al. 2023; Yazıcıoğlu and Akgüngör 2025) have an equal effect on temperate climate where weather-based features have minimal variation.

Hence, the current research gap lies at the intersection of these three dimensions, geographical limitation (lack of temperate climates validation), methodological limitations (absence of systematic ablation study), and analytical limitation (lack of feature importance to identify key drivers). In addressing these gaps, a methodical approach that explores STGNN performance with progressively added feature categories in a tropical climate setting is required.

1.2 Problem Statement

Despite the growing body of research on STGNN for metro ridership prediction, certain gaps are still present. Existing research heavily utilises dataset that are obtained from countries with temperate climates such as China, USA and Europe, overlooking the validation of such research towards countries with tropical climates like Malaysia. The generalisability of these architectures and their feature requirements between temperate and tropical climate remains unexplored. Specifically, certain studies from temperate regions consistently report that temperature and precipitation are important predictors of ridership (Farahmand et al. 2023; Yazıcıoğlu and Akgüngör 2025) , but it is unclear whether these findings hold in tropical contexts where temperature variation is minimal compared to temperate regions.

STGNN with its representation learning has the capacity to learn complex spatio-temporal representations from without any manual feature engineering. Interestingly, some studies have included manually engineered features related to time, weather, holiday and even built-environment (Ren et al. 2025). Nonetheless, certain amongst them did not explicitly and systematically evaluate the contribution of these additional features through ablation study. With the absence of ablation study, it remains a gap in determining the contribution of different feature categories.

Although ablation studies were conducted (Farahmand et al. 2023; Zhang et al. 2024), there is limited understanding of which specific features within temporal and weather categories demonstrate the strongest predictive importance in tropical contexts. Studies from temperate regions have identified temperature as a dominant weather predictor (Farahmand et al. 2023; Yazıcıoğlu and Akgüngör 2025) and peak hours as key temporal indicators (Liu et al. 2025; Zou, Wang, and Guo 2024) but these feature importance may not translate directly to tropical climates. It remains unexplored if tropical-specific temporal patterns and weather patterns such as variance in terms of humidity and precipitation exhibit different predictive importance compared to those in temperate climate.

1.3 Research Questions

Anchoring on the identified research gaps, this study intends to address the following research questions:

RQ1: What is the baseline prediction performance of STGNN using only historical ridership data and spatial network structure for metro ridership in tropical climates?

RQ2: What is the effect towards model performance of STGNN when temporal features (time encodings, holidays, weekends) and weather features (temperature, precipitation, humidity) are integrated on metro data with tropical climate?

RQ3: Which specific temporal and weather features demonstrate the strongest feature importance in tropical climate ridership prediction?

1.4 Research Objectives

RO1: To evaluate performance of baseline STGNN architecture based on standard regression metrics.

RO2: To conduct ablation study on the contribution of temporal features and weather features towards STGNN prediction performance.

RO3: To analyse the feature importance of temporal and weather features through standard model interpretability methods.

1.5 Research Scope and Contribution

This research focuses on short-term ridership prediction, employing a 23-hour historical window to forecast next-hour ridership, specifically targeting Malaysia's KTM Komuter urban rail

transit system. The spatial scope encompasses station-level ridership modeling, where relationships between stations are represented through graph structures. The temporal scope covers hourly, daily, and weekly patterns across both regular periods and special events such as holidays and major public gatherings. The research explicitly excludes other transportation modes including buses, taxis, LRT, and MRT systems, maintaining focus on the KTM Komuter network to ensure methodological rigor and depth of analysis.

The research aims to contribute from both theoretical and practical perspectives. Theoretically, it validates STGNN architecture in tropical climate contexts, addressing the geographical bias in existing literature that predominantly features temperate climate datasets from China, USA, and Europe. Additionally, it provides empirical evidence on the contribution of manual feature integration within representation learning architectures like STGNNs, enhancing understanding of the relationship between explicit manual features and implicit representation learning. From a practical standpoint, the research develops a metro ridership prediction framework that provides transit operators with actionable data-driven insights for resource optimisation. Furthermore, the ablation study methodology is designed to be adaptable to other Southeast Asian countries with tropical climates, offering a replicable approach for evaluating transit system prediction models in similar geographical contexts.

2 Literature Review

2.1 Spatio-Temporal Graph Neural Networks for Transit Prediction

Spatio-Temporal Graph Neural Networks have emerged as powerful architectures for transit ridership prediction by capturing both spatial dependencies leveraging graph structures and temporal dynamics through sequential modeling. Recent architectural innovations demonstrate this evolution, with Li et al. (2025) proposing a multi-dimensional time-dependent dynamic GNN that employs 1D dilated convolutions to extract both short-term and long-term temporal features, while Liu et al. (2025) developed a dynamic spatio-temporal graph fusion network utilising non-negative tensor decomposition for dynamic virtual graphs with parallel learning architecture. These models typically integrate Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) with temporal components such as LSTM, GRU, or Temporal Convolutional Networks (TCN) to model complex spatiotemporal dependencies. Ma et al. (2025) addressed the over-smoothing problem in deep GNNs through a second-order continuous framework with ILN layers, achieving improved performance on traffic flow prediction. Wang et al. (2024) extended STGNN capabilities by introducing probabilistic frameworks that quantify spatiotemporal uncertainty through multi-graph proximity representations, moving beyond deterministic point estimates to provide confidence intervals for predictions. These architectural advances collectively demonstrate the field's progression toward more sophisticated modeling of complex spatiotemporal patterns, though computational costs remain a significant consideration with models processing datasets exceeding 70 million rows (Li et al. 2025).

2.2 Integration of External Features for Transit Prediction

Zeb et al. (2025) demonstrated the value of pandemic-specific feature integration by incorporating COVID-19 case counts and non-pharmaceutical interventions (NPIs) into LSTM models for BRT ridership prediction in Pakistan, achieving strong performance despite unprecedented disruption to normal travel patterns. Similarly, Zhao, Yu, and Zhang (2025) addressed short-term prediction during unplanned events using Random Forest and gradient-boosting decision trees with SMOTE techniques to handle minority event data. Weather features have received particular attention in temperate climate contexts, with Farahmand et al. (2023) systematically examining meteorological parameters (temperature,

precipitation, wind, humidity) for bus ridership in the Netherlands using MLP architectures with hourly aggregation. Zou et al. (2024) extended feature integration to real-time incident scenarios by developing incident matrices combined with external features using ConvLSTM for Beijing metro OD flow prediction. Beyond conventional features, Ren et al. (2025) implemented multimodal approaches by integrating land-use images with 5D built environment indicators through CNN architectures, employing image-based SHAP for interpretability. These studies collectively demonstrate that while feature integration consistently improves model performance, the selection of relevant features depends heavily on the specific prediction context, temporal horizon, and data availability, with limited understanding of which features remain valuable across different geographical and climatic settings.

2.3 Geographical and Climate Context in Transit Prediction

The geographical distribution of STGNN research reveals a significant concentration in temperate climate regions, particularly China and the United States, with limited exploration of tropical contexts where weather patterns differ fundamentally. Existing studies predominantly focus on systems experiencing substantial seasonal variations, such as the Southwest China and Central China Metro (Li et al. 2025), Shanghai and Hangzhou Metro (Liu et al. 2025), Beijing AFC systems (Ren et al. 2025; Zou et al. 2024), and Chicago transit networks (Wang et al. 2024). In these temperate contexts, weather features demonstrate meaningful impacts on ridership patterns due to extreme temperature variations between summer and winter, seasonal precipitation changes, and their associated effects on travel behavior. Farahmand et al. (2023) found that meteorological parameters significantly influenced bus ridership in the Netherlands' temperate maritime climate, while Zeb et al. (2025) examined the Rawalpindi Islamabad Metro in Pakistan's subtropical climate during the unique circumstances of the COVID-19 pandemic, though with primary focus on pandemic-related features rather than systematic weather effects. The limited representation of tropical climate contexts in the literature creates a gap about the generalisability of feature importance findings, particularly regarding if weather variables that prove valuable in temperate regions maintain predictive value in tropical environments with minimal temperature variation and consistent year-round patterns. This geographical concentration in research contexts suggests that current understanding of optimal feature sets for STGNN models may not adequately address the needs of rapidly expanding urban rail networks in tropical regions such as Southeast Asia, where different meteorological characteristics could warrant different modeling approaches and feature selection strategies.

2.4 Critical Analysis

Author(s) & Year	Research Focus	Focus Type	Contribution	Model	Methodology	Dataset (Country)	Prediction	Evaluation Metrics	Limitation
Li et al. (2025)	Multi-dimensional time-dependent dynamic graph neural network	Architectural Innovation	Captures both short-term and long-term temporal features	GCN, GRU, GAT	1D dilated convolutions to extract multi-dimensional temporal dependencies	Southwest China and Central China Metro (China)	Metro Ridership	MAE,RMSE, Accuracy	Computational costly with 70 million rows, lack of external factors (weather, special-event, incidents)
Liu et al. (2025)	Dynamic Spatio-Temporal Graph Fusion Network modeling	Architectural Innovation	Develop non-negative tensor decomposition techniques for dynamic virtual graphs	TCN, GCN, GAT	Parallel learning architecture, Distributed processing for reducing computational costs	Shanghai and Hangzhou Metro (China)	Metro Ridership	MAE, RMSE, MAPE, R^2	Lack of additional features (weather, special-event, incidents), high model complexity
Zeb et al. (2025)	Forecasting public transit ridership amidst COVID-19	Feature Integration	Consideration of unprecedented circumstance of COVID-19, integration of pandemic-related features	LSTM	Deep learning with seasonality, COVID-19 cases and non-pharmaceutical interventions as features	Rawalpindi Islamabad Metro (Pakistan)	BRT Ridership	MAE, RMSE, MAPE	Performance deteriorates for long-term prediction due to uncertainty, limited to specific pandemic context
Ma et al. (2025)	Spatio-temporal traffic flow forecasting based on second-order continuous graph neural network	Architectural Innovation	Addresses over-smoothing in deep GNNs, second-order continuous framework with ILN layers	SCGNN, ILN	Second-order continuous framework for higher-order spatio-temporal dependencies, multiple SCGNN layers integrated	California and Los Angeles traffic data (USA)	Traffic Flow	MAE, RMSE, MAPE	Not suitable for real-time due to time complexity and large model size, limited explainability, high computational cost
Wang et al. (2024)	Uncertainty quantification of spatiotemporal travel demand with probabilistic graph neural networks	Feature Integration	Quantify spatiotemporal uncertainty in travel demand, develop probabilistic framework	GCN, GAT, LSTM	Probabilistic assumptions, multi-graph proximity	Chicago rail, bus and ridesharing data (USA)	Metro, Bus and Rideshare Demand	NLL, Calibration Error, MPIW, PICP, MAE, MAPE	Requires further validation on other context and cities, limited to specific probabilistic distributions

Author(s) & Year	Research Focus	Focus Type	Contribution	Model	Methodology	Dataset (Country)	Prediction	Evaluation Metrics	Limitation
Wu et al. (2025)	LLM4PT: A large language model-based system for public transit prediction	Architectural Innovation	LLM for transit prediction, provides human-readable explanations	LLaMA3.1-8B	Prompt tuning with Chain-of-Thought Spatiotemporal Aligner with learnable parameters	Hangzhou, Nanjing, New York metro data (China, USA)	Metro OD Flow	WAPE, MAPE	Limited public event data, fine-tuning LLMs remains complex
Zou et al. (2024)	Real-time prediction during underground incidents	Feature Integration	Consideration of incident feature in prediction	ConvLSTM	Incident matrices combined with external features	Beijing AFC metro incident data (China)	Metro OD Flow	MMSE, MMAE, MMAPE	Limited to incident scenarios, requires real-time incident data
Ren et al. (2025)	A multimodal deep learning-based direct ridership model for planning-oriented demand prediction	Feature Integration	Integrates land-use images with 5D built environment indicators	CNN	Land-use image generation from multi-source data; image-based SHAP for interpretability	Beijing URT Network (China)	Metro OD Ridership	MAE, RMSE, MAPE, R^2	Limited to planning scenarios, requires high-quality spatial data
Zhao et al. (2022)	Short-term metro ridership prediction during unplanned events	Feature Integration	Addresses prediction under unplanned events	LSTM, Random Forest, gradient-boosting decision trees	SMOTE to handle minority data of unplanned events	Large-scale Metro AFC data (Undisclosed)	Metro Ridership	MAPE, RMSE	Limited to specific unplanned event types
Farahmand et al. (2023)	Predicting bus ridership based on the weather conditions using deep learning algorithms	Feature Integration	Research impact of weather on bus ridership	MLP	MLP with meteorological parameters (temperature, precipitation, wind, humidity), hourly aggregation	Twente Region Bus Network data (Netherlands)	Bus Ridership	MAE, EVS	Limited to short-term prediction, weather forecast uncertainty

Table 1: *Comparative Analysis Table*

The reviewed literature (8 out of 9) demonstrates substantial geographical concentration in temperate climate regions (China, USA, Netherlands) where high variance of annual temperature and distinct seasonal transitions strongly influence travel behavior. However, the applicability of these findings to tropical contexts remains unexplored. Tropical climates exhibit different characteristics of consistent year-round temperatures, absence of winter precipitation patterns, and minimal seasonal behavioral shifts. The assumption that weather features valuable in temperate settings retain predictive power in tropical environments lacks empirical validation, creating uncertainty about optimal feature selection for Southeast Asian urban rail networks. Furthermore, lack of baseline performance benchmarks exist for STGNNs operating exclusively on historical ridership and spatial structure in tropical contexts. This gap directly motivates RQ1.

Despite widespread feature integration across reviewed studies, the literature exhibits a methodological limitation which is the absence of systematic ablation experiments isolating individual feature contributions. Some studies integrate multiple feature categories yet none systematically decompose which specific features drive performance gains versus adding computational overhead without predictive value (Ren et al. 2025; Zeb et al. 2025; Zou et al. 2024). This methodological gap prevents understanding whether marginal improvements from adding weather features justify increased data collection costs and processing complexity, particularly relevant for resource-constrained operational environments. The lack of controlled ablation experiments limits the understanding of the contribution of temporal and weather features, which directly motivates RQ2.

While several studies employ interpretability methods, such as (Wang et al. 2024) through probabilistic frameworks, (Ren et al. 2025) leveraging SHAP, neither explicitly analyse which temporal and weather features demonstrate strongest importance across different climatic contexts. Existing feature importance analyses remain confined to single geographical settings without comparative evaluation. Temperature features exhibiting high importance in Netherlands' maritime climate (Farahmand et al. 2023) may prove statistically insignificant in Malaysia's tropical context where temperature variation is much lower annually. Similarly, indicators valuable in Chinese metros (Ren et al. 2025; Wu et al. 2025) may lack predictive power in equatorial regions without distinct seasons. This analytical gap limits the optimisation of model complexity relative to geographical deployment context and limits understanding of feature transferability, subsequently motivating RQ3.

3 Research Methodology

3.1 Research Design

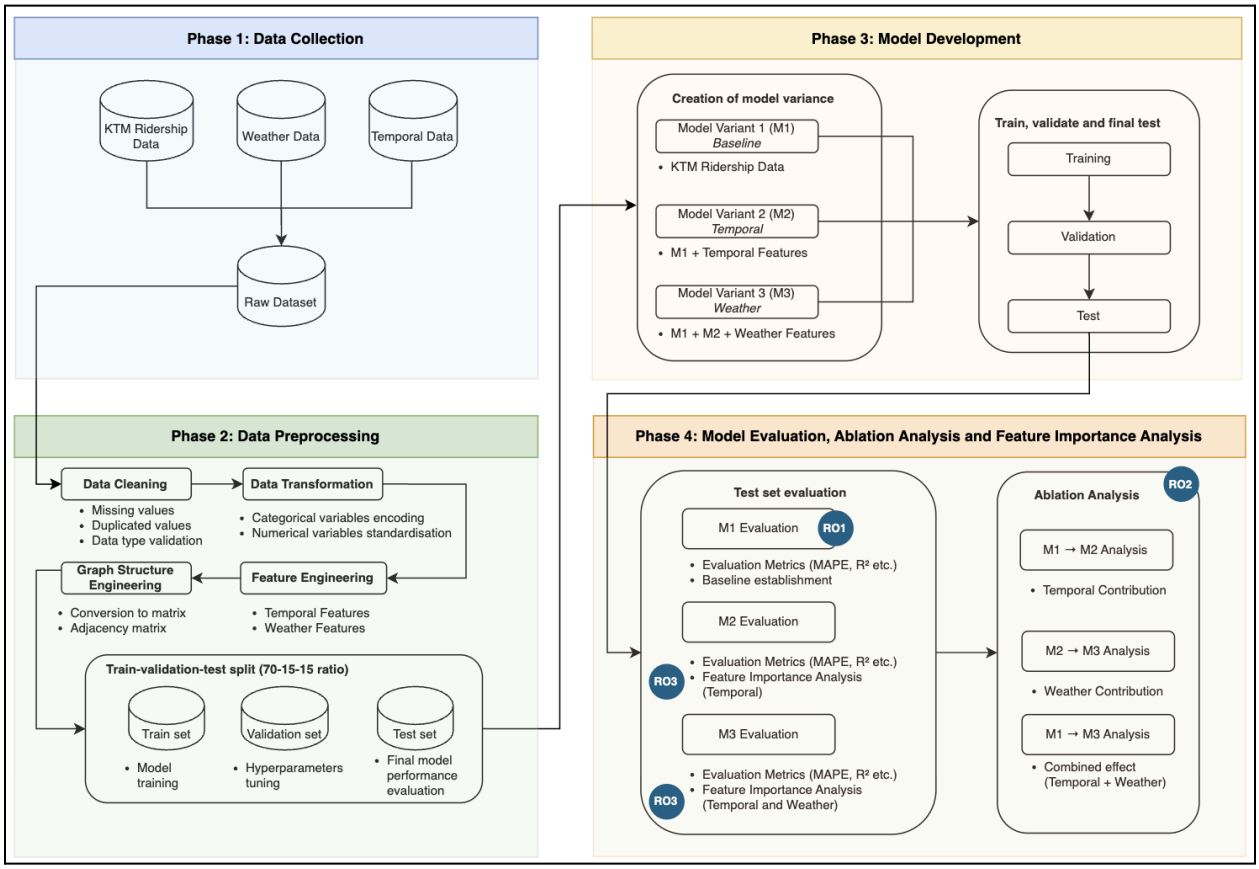

Figure 1: *Four Phases of Research Design*

The research employs an ablation study approach to systematically evaluate the contribution of temporal and weather features, alongside feature importance analysis, in STGNNs for metro ridership prediction in Malaysia's tropical climate context. The research is structured into four sequential phases, which are (P1) data collection, (P2) data preprocessing, (P3) model development, (P4) model evaluation, ablation analysis and feature importance analysis. Data is initially collected and ingested from various sources, with the main one being the KTM ridership data, followed by a sequence of data preprocessing operations including data cleaning, data transformation, feature engineering, data structure engineering and lastly a train-validation-split of 70-15-15 ratio. Next, three model variants are progressively developed, each undergoing training, validation, and testing processes. Subsequently, all variants are evaluated on the test set to establish baseline performance (RO1), ablation analysis compares

variant performance to quantify feature contributions (RO2), while variants M2 and M3 undergo feature importance analysis to identify key predictors (RO3).

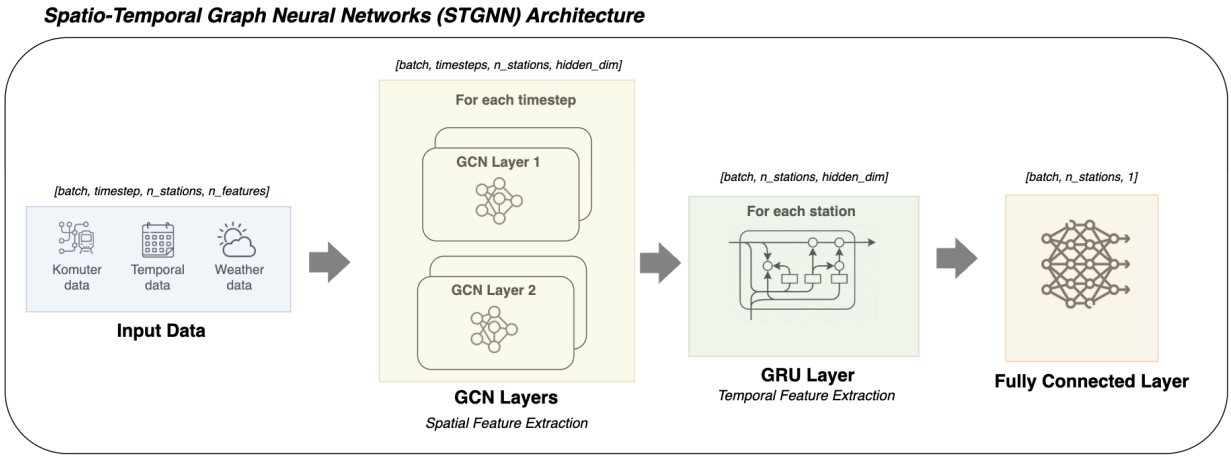

Figure 2: *Architecture for Spatio-Temporal Graph Neural Networks (STGNN)*

3.2 STGNN Model Architecture

The model architecture will be using a sequential stacking architecture combining Graph Convolutional Networks (GCN) and Gated Recurrent Units (GRU) to capture both spatial and temporal dependencies in metro ridership data. Specifically, the architecture consists of two main components:

1. Spatial Module:
   This module captures spatial dependencies between metro stations using the network's topological structure. Two to three **Graph Convolutional Networks (GCN)** will be utilised layers to process the adjacency matrix to learn spatial relationships and propagate information across the metro network.

2. Temporal Module:
   This module captures temporal dependencies in ridership patterns over time. One **Gated Recurrent Units (GRU)** layer processes sequential data to learn temporal dynamics and patterns.

The sequential processing flow operates as follows, firstly the input data passes through the GNN layers to extract spatial features, then the spatially-enriched representations are fed

into GRU layers to model temporal dynamics. This design enables the model to learn both how ridership patterns relate across stations (spatial) and how they evolve over time (temporal). The specific model variants (M1, M2, M3) share this base architecture but differ in their input features as described in Phase 3.

3.3 Phase 1: Data Collection

Phase 1 focuses on gathering three primary data sources required for the research, all obtained through publicly available sources. KTM Komuter ridership data will be collected from Open Data Portal by the Government of Malaysia (Malaysia Open Data Portal n.d.), capturing historical passenger flow patterns across the network mainly for the year 2024. Weather data will be obtained from the Malaysian Meteorological Department (Malaysian Meteorological Department n.d.), including temperature, precipitation, and humidity measurement for the year 2024 as well. Temporal data includes time-related information such as hour of day, day of week, month, and holiday indicators. These three data sources are integrated to form the raw dataset that serves as the foundation for subsequent preprocessing and modeling phases.

The data collection methodology employs a hybrid approach combining automated API calls and manual data extraction. KTM Komuter ridership data will be retrieved programmatically through Python-based API calls to Malaysia's Open Data Portal. Weather data from the Malaysian Meteorological Department will be obtained through manual download, as their data distribution system does not provide API access

3.4 Phase 2: Data Preprocessing

Phase 2 transforms the raw dataset into a format suitable for STGNN modeling through four main steps. Data cleaning addresses missing values, duplicate entries, and validates data types to ensure data quality. Data transformation includes encoding categorical variables and standardising numerical variables to appropriate scales. Feature engineering creates the specific input features for each model variant, temporal features (cyclical time encodings, holiday indicators etc.) and weather features (temperature, precipitation, humidity etc.). Graph structure engineering converts the metro network topology into matrix representations including the adjacency matrix that captures spatial relationships between stations. The preprocessed data is then split into training (70%), validation (15%), and test (15%) sets, where the training

set is used for model training, the validation set for hyperparameter tuning, and the test set reserved exclusively for final model performance evaluation.

All the process above will be implemented using Python and leverage on open access libraries such as pandas for data manipulation, NumPy for numerical operations, scikit-learn for preprocessing and feature encoding, NetworkX for graph structure representation, and PyTorch Geometric for GNN-specific data structures and transformations.

3.5 Phase 3: Model Development

Phase 3 develops three model variants to systematically isolate feature contributions. Model Variant 1 (M1) serves as the baseline, utilising only KTM ridership data and spatial metro network topology. Model Variant 2 (M2) extends M1 by incorporating temporal features, enabling assessment of temporal information's contribution. Model Variant 3 (M3) further extends M2 by adding weather features, allowing evaluation of weather data contribution. The temporal prediction framework utilises a sliding window approach where each model variant ingests 23 consecutive hourly timesteps as input sequences to generate a single-step forecast for the subsequent hour. Each variant undergoes the similar process, training on the training set to learn model parameters, validation on the validation set for hyperparameter optimisation, and final testing on the reserved test set.

The model training implementation will leverage on the free publicly available GPU resource from Google Colab for accelerated training. As for the deep learning framework, PyTorch will be utilised. Specifically, PyTorch Geometric for GCN layers and PyTorch for GRU layers. All hyperparameters and processes will be implemented in manners that enable reproducibility including setting the random seed.

3.6 Phase 4: Model Evaluation, Ablation Analysis and Feature Importance Analysis

Phase 4 serves to evaluate the performance of each model variant, quantify feature contribution through ablation study, and assess feature importance to identify key predictors. The test set evaluation assesses each model variant using standard regression metrics including Mean Absolute Error (MAE) for interpretable average prediction error, Root Mean Squared Error (RMSE) for penalising larger errors, Mean Absolute Percentage Error (MAPE) for

scale-independent comparison, and R^2 for proportion of variance explained to quantify prediction performance (RO1).

The ablation analysis systematically compares model pairs: M1-M2 pair quantifies temporal feature contribution, M2-M3 pair quantifies weather feature contribution, and M1-M3 pair quantifies the combined effect of both feature types (RO2). For variants incorporating additional features (M2 and M3), feature importance analysis employs SHAP (SHapley Additive exPlanations) values to quantify individual feature contributions and permutation importance methods to assess feature impact on model performance, identifying the key predictors that strongly influence predictions (RO3). The implementation utilises the SHAP library for explainability analysis and scikit-learn for permutation importance calculations.

4 Expected Outcome

This study expects to deliver three primary outcomes addressing the identified research gaps. First, baseline STGNN performance benchmarks for tropical climate metro ridership using only historical and spatial data (RO1), establishing an empirical reference for Malaysian context. Second, quantified feature contributions through systematic ablation study comparing M1 (baseline), M2 (additional temporal features), and M3 (additional weather features), measured via standard regression metrics (MAE, RMSE, MAPE, R^2) to determine whether temporal and weather features provide statistically significant improvements in tropical settings (RO2). Third, feature importance rankings identifying which specific temporal (time encodings, holidays) and weather variables (temperature, precipitation, humidity) demonstrate strongest predictive value in tropical climate through SHAP and permutation importance analysis (RO3).

The research may yield few possible scenarios which could all be scientifically valuable, (1) null effects for weather features due to minimal tropical temperature variation, demonstrating non-generalisability and justifying computational optimisation, (2) partial importance for specific weather variables like precipitation while temperature proves insignificant or (3) full replication validating existing models' transferability. Regardless of outcome direction, findings will provide actionable guidance for Malaysian transit operators on optimal feature selection strategies and contribute to SDG 11 (Sustainable Cities and Communities) by improving transit prediction accuracy for better resource allocation, service reliability, and increased public transport

adoption, while supporting SDG 13 (Climate Action) through optimised operations reducing urban carbon emissions.

5 Research Plan

This research will be conducted over 31 weeks from 9 March 2026 to 14 September 2026 under the course *WQD7025 Data Science Research Project*, which spans Semester II (23 weeks) and Special Semester (8 weeks). The research is structured into five phases, Phase 1 (Weeks 1-6) focuses on data collection and preprocessing of KTM Komuter ridership and meteorological data, Phase 2 (Weeks 7-12) involves feature engineering and baseline model development, Phase 3 (Weeks 13-22) covers STGNN architecture design, implementation and training with ablation studies, Phase 4 (Weeks 23-28) encompasses comprehensive model evaluation and comparative analysis; and Phase 5 (Weeks 29-31) is allocated for thesis writing, final submission and presentation. This timeline is feasible within the master's scope with clear deliverables per phase and sufficient time for iterative refinement.

As for the tools and platforms, the research will be implemented using Python with standard data science libraries (NumPy, Pandas, Scikit-learn) for data preprocessing. Deep learning implementation will utilise PyTorch as the core framework and PyTorch Geometric (PyG) for graph neural network development, while model training will leverage publicly available GPU resources from Google Colab to ensure efficient and accessible computation for STGNN training and experimentation.

References

Farahmand, Zakir H., Konstantinos Gkiotsalitis, and Karst T. Geurs. 2023. 'Predicting Bus Ridership Based on the Weather Conditions Using Deep Learning Algorithms'. *Transportation Research Interdisciplinary Perspectives* 19:100833. doi:10.1016/j.trip.2023.100833.

Jin, Guangyin, Yuxuan Liang, Yuchen Fang, Zezhi Shao, Jincai Huang, Junbo Zhang, and Yu Zheng. 2024. 'Spatio-Temporal Graph Neural Networks for Predictive Learning in Urban Computing: A Survey'. *IEEE Transactions on Knowledge and Data Engineering* 36(10):5388–5408. doi:10.1109/TKDE.2023.3333824.

Li, Ruisen, Liqiang Zhao, Jinjin Tang, Shuixiong Tang, and Zhenxing Hao. 2025. 'Multi-Dimensional Time-Dependent Dynamic Graph Neural Network for Metro Passenger Flow Prediction'. *Applied Intelligence* 55(6):442. doi:10.1007/s10489-025-06346-z.

Liu, Wenzheng, Hongtao Li, Haina Zhang, Jiang Xue, and Shaolong Sun. 2025a. 'Dynamic Spatio-Temporal Graph Fusion Network Modeling for Urban Metro Ridership Prediction'. *Information Fusion* 117:102845. doi:10.1016/j.inffus.2024.102845.

Ma, Zhaobin, Zhiqiang Lv, Zhihao Xu, Rongkun Ye, and Jianbo Li. 2025. 'Spatio-Temporal Traffic Flow Forecasting Based on Second-Order Continuous Graph Neural Network'. *The Computer Journal* 68(9):1300–1314. doi:10.1093/comjnl/bxaf038.

Malaysia Open Data Portal, Government of. n.d. 'Hourly Origin-Destination Ridership: Komuter | Data.Gov.My'. Retrieved 11 January 2026. https://data.gov.my/data-catalogue/ridership_od_komuter.

Malaysian Meteorological Department. n.d. 'myMETdata | Home'. Retrieved 11 January 2026. https://mymetdata.met.gov.my/.

Ren, Peng, Shuang Liu, Beining Qin, Yue Chen, Qi Xu, and Peng He. 2025. 'A Novel Multimodal Deep Learning-Based Direct Ridership Model for Planning-Oriented Demand Prediction in Urban Rail Transit Networks'. *Journal of Transport Geography* 129:104409. doi:10.1016/j.jtrangeo.2025.104409.

Wang, Qingyi, Shenhao Wang, Dingyi Zhuang, Haris Koutsopoulos, and Jinhua Zhao. 2024. 'Uncertainty Quantification of Spatiotemporal Travel Demand With Probabilistic Graph Neural Networks'. *IEEE Transactions on Intelligent Transportation Systems* 25(8):8770–81. doi:10.1109/TITS.2024.3367779.

Wu, Fei, Changjiang Zheng, Muqing Du, Shukang Zheng, Junze Ma, and Ye Lu. 2025. 'LLM4PT: A Large Language Model-Based System for Flexible and Explainable Public Transit Demand Prediction'. *Computers & Industrial Engineering* 209:111449. doi:10.1016/j.cie.2025.111449.

Yazıcıoğlu, Cafer, and Ali Akgüngör. 2025. 'Modeling Short-Term Passenger Flows in Metro and Bus Systems Using Meteorological Data: Deep Learning Model Comparisons'. *Applied Sciences* 15(11):6260. doi:10.3390/app15116260.

Zeb, Muhammad Shah, Muhammad Asif Khan, Muhammad Muzzamil Hussain Khattak, Sameer Ud-Din, Muhammad Faisal Habib, and Muhammad Zaheer Khan. 2025. 'Forecasting Public Transit Ridership amidst COVID-19: A Machine Learning Approach'. *Public Transport* 17(2):391–420. doi:10.1007/s12469-024-00368-5.

Zhang, Shuxin, Jinlei Zhang, Lixing Yang, Feng Chen, Shukai Li, and Ziyou Gao. 2024. 'Physics Guided Deep Learning-Based Model for Short-Term Origin–Destination Demand Prediction in Urban Rail Transit Systems Under Pandemic'. *Engineering* 41:276–96. doi:10.1016/j.eng.2024.04.020.

Zhao, Xinru, Wenhao Yu, and Yifan Zhang. 2025. 'MST-GNN: Graph Neural Network with Multi-Granularity in Space and Time for Traffic Prediction'. *GeoInformatica* 29(3):377–401. doi:10.1007/s10707-024-00532-w.

Zou, Linmu, Zijia Wang, and Rui Guo. 2024. 'Real-Time Prediction of Transit Origin–Destination Flows during Underground Incidents'. *Transportation Research Part C: Emerging Technologies* 163:104622. doi:10.1016/j.trc.2024.104622.

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WOX7001 RESEARCH METHODOLOGY 1/2025/2026

Temporal and Weather Feature Integration for Spatio-Temporal Graph Neural Networks Metro Ridership Prediction in Malaysia's Tropical Climate

40

RESEARCH PROPOSAL

Matric Number:	24087980
Name	Faiz Yah Bin Abdul Hanan Yah
Occurrence:	7
Lecturer Name:	Assoc. Prof. Dr. Norjihan Abdul Ghani

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Abstract

Urban rail transit systems are expanding rapidly in tropical Southeast Asian cities which calls for accurate ridership prediction models for effective operational planning and resource allocation. Spatio-Temporal Graph Neural Networks (STGNNs) have demonstrated promising performance for metro ridership prediction, yet existing research exhibits substantial geographical concentration in temperate climate regions (China, USA, Netherlands) where weather shows higher variability. The generalisability of temporal and weather features to tropical contexts remains empirically unvalidated, while the literature lacks systematic ablation studies isolating individual feature contributions and feature importance analysis. This research addresses these gaps through a three-variant ablation study evaluating STGNN performance in Malaysia's tropical climate, baseline model using only historical ridership and spatial structure (M1), baseline with temporal features (M2), and baseline with temporal and weather features (M3). The architecture of STGNN is a combination of GCN spatial architecture with GRU temporal architecture, utilising a 23-hour lookback window to predict next-hour, modelled on KTM Komuter data 2024. The study quantifies incremental contributions of each feature category through standard regression metrics (MAE, RMSE, MAPE, R^2) and identifies strongest predictors via SHAP and permutation importance methods. The methodology is tested on KTM Komuter annual ridership data from 2024, encompassing station-level transactions, meteorological variables, and Malaysian holiday calendars. The findings are expected to provide empirical validation of temperate-climate assumptions in tropical settings, quantified guidance on optimal feature selection for computational efficiency, and climate-specific importance rankings. This research contributes to understanding STGNN generalisability across diverse geographical contexts while providing actionable insights for transit operators on cost-effective model deployment, supporting SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action) through improved prediction accuracy and optimised operations.

Keywords: Spatio-Temporal Graph Neural Networks, Metro Ridership Prediction, Tropical Climate, Feature Importance Analysis, Ablation Study

1 Introduction

1.1 Research Background

Urban rail transit systems play a critical role in ensuring mobility of the population, especially in densely populated metropolitan areas. An accurate metro ridership prediction is vital for optimisation of resource allocation, improvement of service quality and enhancement of operational efficiency. Recent advances in deep learning, particularly Graph Neural Networks (GNNs) have shown promising performance in capturing spatio-temporal dependencies inherent in metro ridership (Jin et al. 2024; Li et al. 2025; Wang et al. 2024), since the connections between stations represent the spatial pattern while the hourly ridership reflects the temporal patterns.

Spatio-Temporal Graph Neural Networks (STGNNs) have emerged as a strong architecture for transit ridership prediction with its capacity in capturing the relationship of inherent spatial nature of metro connections combined with the temporal dynamics of the hourly ridership. Short-term ridership prediction for metro systems typically uses historical data from 12 to 24 hours to forecast demand 10 to 60 minutes ahead, enabling effective operational planning and real-time resource allocation (Jin et al. 2024). Studies such as Multi-dimensional Time-dependent Dynamic Graph Neural Network, (MTDGNN) (Li et al. 2025), Dynamic Spatio-Temporal Graph Fusion Network (DSTGFN) (Liu et al. 2025) have demonstrated improved performance compared to traditional time series methods. These models leverage graph structure of metro networks to learn spatial dependencies and integrate temporal modules to learn the ridership patterns across time.

However, there exists several gaps in the current body of literature. Firstly, a large majority of STGNN research has been using datasets from regions with temperate climate, predominantly in China and the United States. Cities in these countries such as Beijing, Shanghai and New York have significant temperature changes across the year which has an impact on the ridership patterns. Secondly, while some studies have incorporated external features such as weather and temporal context (Farahmand, Gkiotsalitis, and Geurs 2023), there remain gaps in systematic analysis on how features contribute to prediction performance. Thirdly, even if ablation study was implemented, there is limited empirical evidence on whether similar features that contribute in temperate climates remain relevant in tropical contexts.

As the majority of research was conducted with temperate climate dataset, Malaysia poses an unique opportunity for validating the STGNN model in context with tropical climate characteristics. Unlike temperate regions with drastic temperature changes, Malaysia has a relatively stable temperature throughout the year, high humidity and distinct rainfall patterns. It is crucial to then identify if weather-based features that improve prediction in temperate climates (Farahmand et al. 2023; Yazıcıoğlu and Akgüngör 2025) have an equal effect on temperate climate where weather-based features have minimal variation.

Hence, the current research gap lies at the intersection of these three dimensions, geographical limitation (lack of temperate climates validation), methodological limitations (absence of systematic ablation study), and analytical limitation (lack of feature importance to identify key drivers). In addressing these gaps, a methodical approach that explores STGNN performance with progressively added feature categories in a tropical climate setting is required.

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1.2 Problem Statement

Despite the growing body of research on STGNN for metro ridership prediction, certain gaps are still present. Existing research heavily utilises dataset that are obtained from countries with temperate climates such as China, USA and Europe, overlooking the validation of such research towards countries with tropical climates like Malaysia. The generalisability of these architectures and their feature requirements between temperate and tropical climate remains unexplored. Specifically, certain studies from temperate regions consistently report that temperature and precipitation are important predictors of ridership (Farahmand et al. 2023; Yazıcıoğlu and Akgüngör 2025) , but it is unclear whether these findings hold in tropical contexts where temperature variation is minimal compared to temperate regions.

STGNN with its representation learning has the capacity to learn complex spatio-temporal representations from without any manual feature engineering. Interestingly, some studies have included manually engineered features related to time, weather, holiday and even built-environment (Ren et al. 2025). Nonetheless, certain amongst them did not explicitly and systematically evaluate the contribution of these additional features through ablation study. With the absence of ablation study, it remains a gap in determining the contribution of different feature categories.

Although ablation studies were conducted (Farahmand et al. 2023; Zhang et al. 2024), there is limited understanding of which specific features within temporal and weather categories demonstrate the strongest predictive importance in tropical contexts. Studies from temperate regions have identified temperature as a dominant weather predictor (Farahmand et al. 2023; Yazıcıoğlu and Akgüngör 2025) and peak hours as key temporal indicators (Liu et al. 2025; Zou, Wang, and Guo 2024) but these feature importance may not translate directly to tropical climates. It remains unexplored if tropical-specific temporal patterns and weather patterns such as variance in terms of humidity and precipitation exhibit different predictive importance compared to those in temperate climate.

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1.3 Research Questions

Anchoring on the identified research gaps, this study intends to address the following research questions:

RQ1: What is the baseline prediction performance of STGNN using only historical ridership data and spatial network structure for metro ridership in tropical climates?

RQ2: What is the effect towards model performance of STGNN when temporal features (time encodings, holidays, weekends) and weather features (temperature, precipitation, humidity) are integrated on metro data with tropical climate?

RQ3: Which specific temporal and weather features demonstrate the strongest feature importance in tropical climate ridership prediction?

1.4 Research Objectives

RO1: To evaluate performance of baseline STGNN architecture based on standard regression metrics.

RO2: To conduct ablation study on the contribution of temporal features and weather features towards STGNN prediction performance.

RO3: To analyse the feature importance of temporal and weather features through standard model interpretability methods.

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1.5 Research Scope and Contribution

This research focuses on short-term ridership prediction, employing a 23-hour historical window to forecast next-hour ridership, specifically targeting Malaysia's KTM Komuter urban rail

transit system. The spatial scope encompasses station-level ridership modeling, where relationships between stations are represented through graph structures. The temporal scope covers hourly, daily, and weekly patterns across both regular periods and special events such as holidays and major public gatherings. The research explicitly excludes other transportation modes including buses, taxis, LRT, and MRT systems, maintaining focus on the KTM Komuter network to ensure methodological rigor and depth of analysis.

The research aims to contribute from both theoretical and practical perspectives. Theoretically, it validates STGNN architecture in tropical climate contexts, addressing the geographical bias in existing literature that predominantly features temperate climate datasets from China, USA, and Europe. Additionally, it provides empirical evidence on the contribution of manual feature integration within representation learning architectures like STGNNs, enhancing understanding of the relationship between explicit manual features and implicit representation learning. From a practical standpoint, the research develops a metro ridership prediction framework that provides transit operators with actionable data-driven insights for resource optimisation. Furthermore, the ablation study methodology is designed to be adaptable to other Southeast Asian countries with tropical climates, offering a replicable approach for evaluating transit system prediction models in similar geographical contexts.

2 Literature Review

2.1 Spatio-Temporal Graph Neural Networks for Transit Prediction

Spatio-Temporal Graph Neural Networks have emerged as powerful architectures for transit ridership prediction by capturing both spatial dependencies leveraging graph structures and temporal dynamics through sequential modeling. Recent architectural innovations demonstrate this evolution, with Li et al. (2025) proposing a multi-dimensional time-dependent dynamic GNN that employs 1D dilated convolutions to extract both short-term and long-term temporal features, while Liu et al. (2025) developed a dynamic spatio-temporal graph fusion network utilising non-negative tensor decomposition for dynamic virtual graphs with parallel learning architecture. These models typically integrate Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) with temporal components such as LSTM, GRU, or Temporal Convolutional Networks (TCN) to model complex spatiotemporal dependencies. Ma et al. (2025) addressed the over-smoothing problem in deep GNNs through a second-order continuous framework with ILN layers, achieving improved performance on traffic flow prediction. Wang et al. (2024) extended STGNN capabilities by introducing probabilistic frameworks that quantify spatiotemporal uncertainty through multi-graph proximity representations, moving beyond deterministic point estimates to provide confidence intervals for predictions. These architectural advances collectively demonstrate the field's progression toward more sophisticated modeling of complex spatiotemporal patterns, though computational costs remain a significant consideration with models processing datasets exceeding 70 million rows (Li et al. 2025).

2.2 Integration of External Features for Transit Prediction

Zeb et al. (2025) demonstrated the value of pandemic-specific feature integration by incorporating COVID-19 case counts and non-pharmaceutical interventions (NPIs) into LSTM models for BRT ridership prediction in Pakistan, achieving strong performance despite unprecedented disruption to normal travel patterns. Similarly, Zhao, Yu, and Zhang (2025) addressed short-term prediction during unplanned events using Random Forest and gradient-boosting decision trees with SMOTE techniques to handle minority event data. Weather features have received particular attention in temperate climate contexts, with Farahmand et al. (2023) systematically examining meteorological parameters (temperature,

precipitation, wind, humidity) for bus ridership in the Netherlands using MLP architectures with hourly aggregation. Zou et al. (2024) extended feature integration to real-time incident scenarios by developing incident matrices combined with external features using ConvLSTM for Beijing metro OD flow prediction. Beyond conventional features, Ren et al. (2025) implemented multimodal approaches by integrating land-use images with 5D built environment indicators through CNN architectures, employing image-based SHAP for interpretability. These studies collectively demonstrate that while feature integration consistently improves model performance, the selection of relevant features depends heavily on the specific prediction context, temporal horizon, and data availability, with limited understanding of which features remain valuable across different geographical and climatic settings.

2.3 Geographical and Climate Context in Transit Prediction

The geographical distribution of STGNN research reveals a significant concentration in temperate climate regions, particularly China and the United States, with limited exploration of tropical contexts where weather patterns differ fundamentally. Existing studies predominantly focus on systems experiencing substantial seasonal variations, such as the Southwest China and Central China Metro (Li et al. 2025), Shanghai and Hangzhou Metro (Liu et al. 2025), Beijing AFC systems (Ren et al. 2025; Zou et al. 2024), and Chicago transit networks (Wang et al. 2024). In these temperate contexts, weather features demonstrate meaningful impacts on ridership patterns due to extreme temperature variations between summer and winter, seasonal precipitation changes, and their associated effects on travel behavior. Farahmand et al. (2023) found that meteorological parameters significantly influenced bus ridership in the Netherlands' temperate maritime climate, while Zeb et al. (2025) examined the Rawalpindi Islamabad Metro in Pakistan's subtropical climate during the unique circumstances of the COVID-19 pandemic, though with primary focus on pandemic-related features rather than systematic weather effects. The limited representation of tropical climate contexts in the literature creates a gap about the generalisability of feature importance findings, particularly regarding if weather variables that prove valuable in temperate regions maintain predictive value in tropical environments with minimal temperature variation and consistent year-round patterns. This geographical concentration in research contexts suggests that current understanding of optimal feature sets for STGNN models may not adequately address the needs of rapidly expanding urban rail networks in tropical regions such as Southeast Asia, where different meteorological characteristics could warrant different modeling approaches and feature selection strategies.

2.4 Critical Analysis

Author(s) & Year	Research Focus	Focus Type	Contribution	Model	Methodology	Dataset (Country)	Prediction	Evaluation Metrics	Limitation
Li et al. (2025)	Multi-dimensional time-dependent dynamic graph neural network	Architectural Innovation	Captures both short-term and long-term temporal features	GCN, GRU, GAT	1D dilated convolutions to extract multi-dimensional temporal dependencies	Southwest China and Central China Metro (China)	Metro Ridership	MAE,RMSE, Accuracy	Computational costly with 70 million rows, lack of external factors (weather, special-event, incidents)
Liu et al. (2025)	Dynamic Spatio-Temporal Graph Fusion Network modeling	Architectural Innovation	Develop non-negative tensor decomposition techniques for dynamic virtual graphs	TCN, GCN, GAT	Parallel learning architecture, Distributed processing for reducing computational costs	Shanghai and Hangzhou Metro (China)	Metro Ridership	MAE, RMSE, MAPE, R^2	Lack of additional features (weather, special-event, incidents), high model complexity
Zeb et al. (2025)	Forecasting public transit ridership amidst COVID-19	Feature Integration	Consideration of unprecedented circumstance of COVID-19, integration of pandemic-related features	LSTM	Deep learning with seasonality, COVID-19 cases and non-pharmaceutical interventions as features	Rawalpindi Islamabad Metro (Pakistan)	BRT Ridership	MAE, RMSE, MAPE	Performance deteriorates for long-term prediction due to uncertainty, limited to specific pandemic context
Ma et al. (2025)	Spatio-temporal traffic flow forecasting based on second-order continuous graph neural network	Architectural Innovation	Addresses over-smoothing in deep GNNs, second-order continuous framework with ILN layers	SCGNN, ILN	Second-order continuous framework for higher-order spatio-temporal dependencies, multiple SCGNN layers integrated	California and Los Angeles traffic data (USA)	Traffic Flow	MAE, RMSE, MAPE	Not suitable for real-time due to time complexity and large model size, limited explainability, high computational cost
Wang et al. (2024)	Uncertainty quantification of spatiotemporal travel demand with probabilistic graph neural networks	Feature Integration	Quantify spatiotemporal uncertainty in travel demand, develop probabilistic framework	GCN, GAT, LSTM	Probabilistic assumptions, multi-graph proximity	Chicago rail, bus and ridesharing data (USA)	Metro, Bus and Rideshare Demand	NLL, Calibration Error, MPIW, PICP, MAE, MAPE	Requires further validation on other context and cities, limited to specific probabilistic distributions

Author(s) & Year	Research Focus	Focus Type	Contribution	Model	Methodology	Dataset (Country)	Prediction	Evaluation Metrics	Limitation
Wu et al. (2025)	LLM4PT: A large language model-based system for public transit prediction	Architectural Innovation	LLM for transit prediction, provides human-readable explanations	LLaMA3.1-8B	Prompt tuning with Chain-of-Thought Spatiotemporal Aligner with learnable parameters	Hangzhou, Nanjing, New York metro data (China, USA)	Metro OD Flow	WAPE, MAPE	Limited public event data, fine-tuning LLMs remains complex
Zou et al. (2024)	Real-time prediction during underground incidents	Feature Integration	Consideration of incident feature in prediction	ConvLSTM	Incident matrices combined with external features	Beijing AFC metro incident data (China)	Metro OD Flow	MMSE, MMAE, MMAPE	Limited to incident scenarios, requires real-time incident data
Ren et al. (2025)	A multimodal deep learning-based direct ridership model for planning-oriented demand prediction	Feature Integration	Integrates land-use images with 5D built environment indicators	CNN	Land-use image generation from multi-source data; image-based SHAP for interpretability	Beijing URT Network (China)	Metro OD Ridership	MAE, RMSE, MAPE, R^2	Limited to planning scenarios, requires high-quality spatial data
Zhao et al. (2022)	Short-term metro ridership prediction during unplanned events	Feature Integration	Addresses prediction under unplanned events	LSTM, Random Forest, gradient-boosting decision trees	SMOTE to handle minority data of unplanned events	Large-scale Metro AFC data (Undisclosed)	Metro Ridership	MAPE, RMSE	Limited to specific unplanned event types
Farahmand et al. (2023)	Predicting bus ridership based on the weather conditions using deep learning algorithms	Feature Integration	Research impact of weather on bus ridership	MLP	MLP with meteorological parameters (temperature, precipitation, wind, humidity), hourly aggregation	Twente Region Bus Network data (Netherlands)	Bus Ridership	MAE, EVS	Limited to short-term prediction, weather forecast uncertainty

Table 1: *Comparative Analysis Table*

The reviewed literature (8 out of 9) demonstrates substantial geographical concentration in temperate climate regions (China, USA, Netherlands) where high variance of annual temperature and distinct seasonal transitions strongly influence travel behavior. However, the applicability of these findings to tropical contexts remains unexplored. Tropical climates exhibit different characteristics of consistent year-round temperatures, absence of winter precipitation patterns, and minimal seasonal behavioral shifts. The assumption that weather features valuable in temperate settings retain predictive power in tropical environments lacks empirical validation, creating uncertainty about optimal feature selection for Southeast Asian urban rail networks. Furthermore, lack of baseline performance benchmarks exist for STGNNs operating exclusively on historical ridership and spatial structure in tropical contexts. This gap directly motivates RQ1.

Despite widespread feature integration across reviewed studies, the literature exhibits a methodological limitation which is the absence of systematic ablation experiments isolating individual feature contributions. Some studies integrate multiple feature categories yet none systematically decompose which specific features drive performance gains versus adding computational overhead without predictive value (Ren et al. 2025; Zeb et al. 2025; Zou et al. 2024). This methodological gap prevents understanding whether marginal improvements from adding weather features justify increased data collection costs and processing complexity, particularly relevant for resource-constrained operational environments. The lack of controlled ablation experiments limits the understanding of the contribution of temporal and weather features, which directly motivates RQ2.

While several studies employ interpretability methods, such as (Wang et al. 2024) through probabilistic frameworks, (Ren et al. 2025) leveraging SHAP, neither explicitly analyse which temporal and weather features demonstrate strongest importance across different climatic contexts. Existing feature importance analyses remain confined to single geographical settings without comparative evaluation. Temperature features exhibiting high importance in Netherlands' maritime climate (Farahmand et al. 2023) may prove statistically insignificant in Malaysia's tropical context where temperature variation is much lower annually. Similarly, indicators valuable in Chinese metros (Ren et al. 2025; Wu et al. 2025) may lack predictive power in equatorial regions without distinct seasons. This analytical gap limits the optimisation of model complexity relative to geographical deployment context and limits understanding of feature transferability, subsequently motivating RQ3.

3 Research Methodology

3.1 Research Design

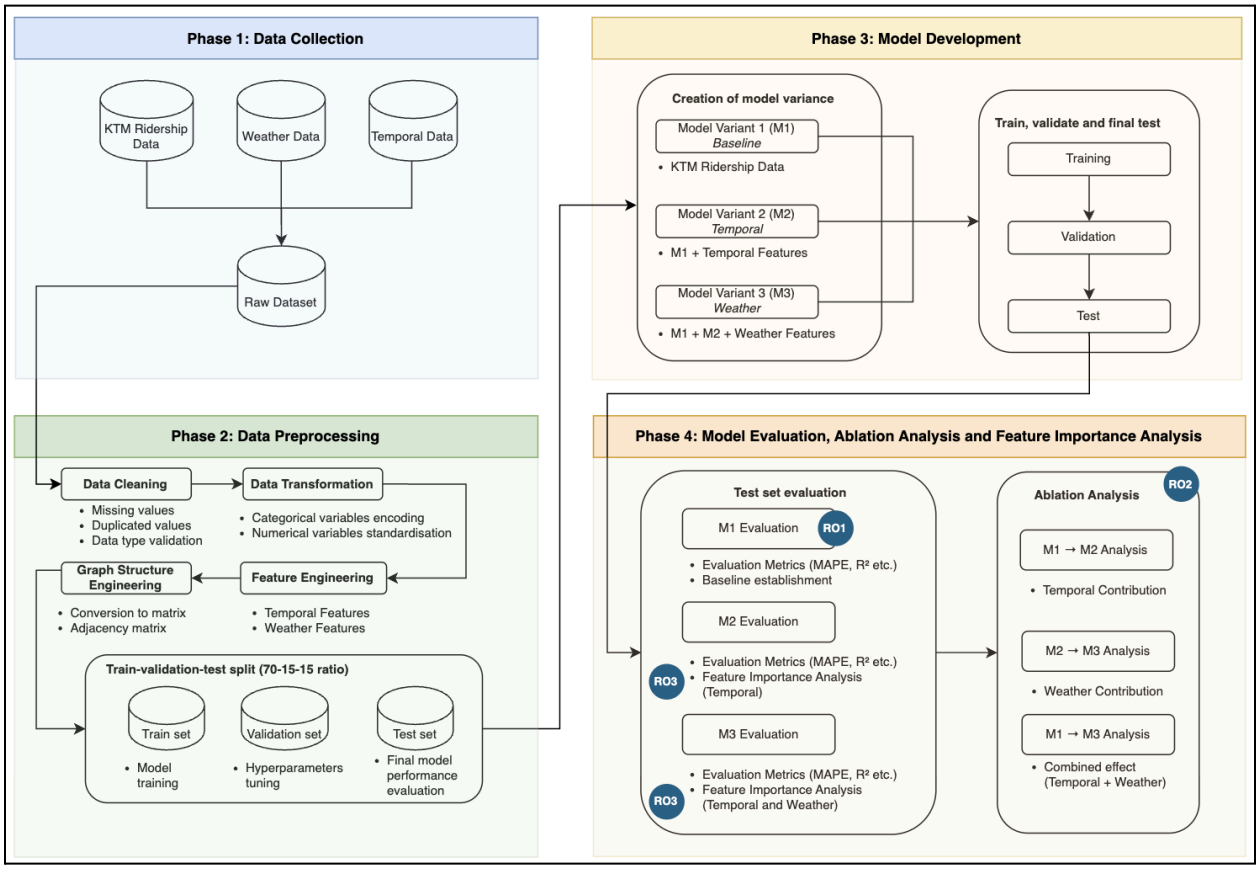

Figure 1: *Four Phases of Research Design*

The research employs an ablation study approach to systematically evaluate the contribution of temporal and weather features, alongside feature importance analysis, in STGNNs for metro ridership prediction in Malaysia's tropical climate context. The research is structured into four sequential phases, which are (P1) data collection, (P2) data preprocessing, (P3) model development, (P4) model evaluation, ablation analysis and feature importance analysis. Data is initially collected and ingested from various sources, with the main one being the KTM ridership data, followed by a sequence of data preprocessing operations including data cleaning, data transformation, feature engineering, data structure engineering and lastly a train-validation-split of 70-15-15 ratio. Next, three model variants are progressively developed, each undergoing training, validation, and testing processes. Subsequently, all variants are evaluated on the test set to establish baseline performance (RO1), ablation analysis compares

variant performance to quantify feature contributions (RO2), while variants M2 and M3 undergo feature importance analysis to identify key predictors (RO3).

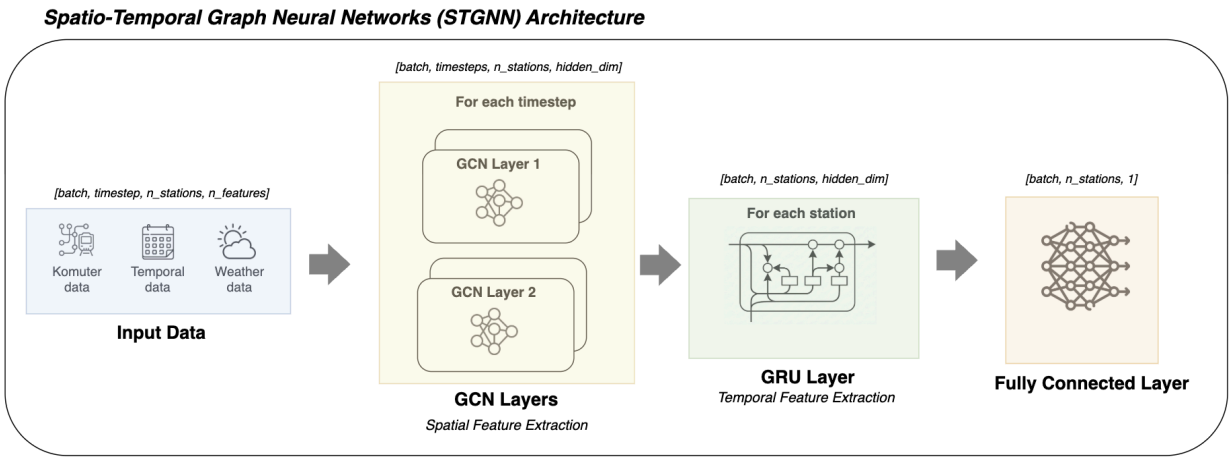

Figure 2: *Architecture for Spatio-Temporal Graph Neural Networks (STGNN)*

3.2 STGNN Model Architecture

The model architecture will be using a sequential stacking architecture combining Graph Convolutional Networks (GCN) and Gated Recurrent Units (GRU) to capture both spatial and temporal dependencies in metro ridership data. Specifically, the architecture consists of two main components:

1. Spatial Module:
   This module captures spatial dependencies between metro stations using the network's topological structure. Two to three **Graph Convolutional Networks (GCN)** will be utilised layers to process the adjacency matrix to learn spatial relationships and propagate information across the metro network.

2. Temporal Module:
   This module captures temporal dependencies in ridership patterns over time. One **Gated Recurrent Units (GRU)** layer processes sequential data to learn temporal dynamics and patterns.

The sequential processing flow operates as follows, firstly the input data passes through the GNN layers to extract spatial features, then the spatially-enriched representations are fed

into GRU layers to model temporal dynamics. This design enables the model to learn both how ridership patterns relate across stations (spatial) and how they evolve over time (temporal). The specific model variants (M1, M2, M3) share this base architecture but differ in their input features as described in Phase 3.

3.3 Phase 1: Data Collection

Phase 1 focuses on gathering three primary data sources required for the research, all obtained through publicly available sources. KTM Komuter ridership data will be collected from Open Data Portal by the Government of Malaysia (Malaysia Open Data Portal n.d.), capturing historical passenger flow patterns across the network mainly for the year 2024. Weather data will be obtained from the Malaysian Meteorological Department (Malaysian Meteorological Department n.d.), including temperature, precipitation, and humidity measurement for the year 2024 as well. Temporal data includes time-related information such as hour of day, day of week, month, and holiday indicators. These three data sources are integrated to form the raw dataset that serves as the foundation for subsequent preprocessing and modeling phases.

The data collection methodology employs a hybrid approach combining automated API calls and manual data extraction. KTM Komuter ridership data will be retrieved programmatically through Python-based API calls to Malaysia's Open Data Portal. Weather data from the Malaysian Meteorological Department will be obtained through manual download, as their data distribution system does not provide API access

3.4 Phase 2: Data Preprocessing

Phase 2 transforms the raw dataset into a format suitable for STGNN modeling through four main steps. Data cleaning addresses missing values, duplicate entries, and validates data types to ensure data quality. Data transformation includes encoding categorical variables and standardising numerical variables to appropriate scales. Feature engineering creates the specific input features for each model variant, temporal features (cyclical time encodings, holiday indicators etc.) and weather features (temperature, precipitation, humidity etc.). Graph structure engineering converts the metro network topology into matrix representations including the adjacency matrix that captures spatial relationships between stations. The preprocessed data is then split into training (70%), validation (15%), and test (15%) sets, where the training

set is used for model training, the validation set for hyperparameter tuning, and the test set reserved exclusively for final model performance evaluation.

All the process above will be implemented using Python and leverage on open access libraries such as pandas for data manipulation, NumPy for numerical operations, scikit-learn for preprocessing and feature encoding, NetworkX for graph structure representation, and PyTorch Geometric for GNN-specific data structures and transformations.

3.5 Phase 3: Model Development

Phase 3 develops three model variants to systematically isolate feature contributions. Model Variant 1 (M1) serves as the baseline, utilising only KTM ridership data and spatial metro network topology. Model Variant 2 (M2) extends M1 by incorporating temporal features, enabling assessment of temporal information's contribution. Model Variant 3 (M3) further extends M2 by adding weather features, allowing evaluation of weather data contribution. The temporal prediction framework utilises a sliding window approach where each model variant ingests 23 consecutive hourly timesteps as input sequences to generate a single-step forecast for the subsequent hour. Each variant undergoes the similar process, training on the training set to learn model parameters, validation on the validation set for hyperparameter optimisation, and final testing on the reserved test set.

The model training implementation will leverage on the free publicly available GPU resource from Google Colab for accelerated training. As for the deep learning framework, PyTorch will be utilised. Specifically, PyTorch Geometric for GCN layers and PyTorch for GRU layers. All hyperparameters and processes will be implemented in manners that enable reproducibility including setting the random seed.

3.6 Phase 4: Model Evaluation, Ablation Analysis and Feature Importance Analysis

Phase 4 serves to evaluate the performance of each model variant, quantify feature contribution through ablation study, and assess feature importance to identify key predictors. The test set evaluation assesses each model variant using standard regression metrics including Mean Absolute Error (MAE) for interpretable average prediction error, Root Mean Squared Error (RMSE) for penalising larger errors, Mean Absolute Percentage Error (MAPE) for

scale-independent comparison, and R^2 for proportion of variance explained to quantify prediction performance (RO1).

The ablation analysis systematically compares model pairs: M1-M2 pair quantifies temporal feature contribution, M2-M3 pair quantifies weather feature contribution, and M1-M3 pair quantifies the combined effect of both feature types (RO2). For variants incorporating additional features (M2 and M3), feature importance analysis employs SHAP (SHapley Additive exPlanations) values to quantify individual feature contributions and permutation importance methods to assess feature impact on model performance, identifying the key predictors that strongly influence predictions (RO3). The implementation utilises the SHAP library for explainability analysis and scikit-learn for permutation importance calculations.

4 Expected Outcome

This study expects to deliver three primary outcomes addressing the identified research gaps. First, baseline STGNN performance benchmarks for tropical climate metro ridership using only historical and spatial data (RO1), establishing an empirical reference for Malaysian context. Second, quantified feature contributions through systematic ablation study comparing M1 (baseline), M2 (additional temporal features), and M3 (additional weather features), measured via standard regression metrics (MAE, RMSE, MAPE, R^2) to determine whether temporal and weather features provide statistically significant improvements in tropical settings (RO2). Third, feature importance rankings identifying which specific temporal (time encodings, holidays) and weather variables (temperature, precipitation, humidity) demonstrate strongest predictive value in tropical climate through SHAP and permutation importance analysis (RO3).

The research may yield few possible scenarios which could all be scientifically valuable, (1) null effects for weather features due to minimal tropical temperature variation, demonstrating non-generalisability and justifying computational optimisation, (2) partial importance for specific weather variables like precipitation while temperature proves insignificant or (3) full replication validating existing models' transferability. Regardless of outcome direction, findings will provide actionable guidance for Malaysian transit operators on optimal feature selection strategies and contribute to SDG 11 (Sustainable Cities and Communities) by improving transit prediction accuracy for better resource allocation, service reliability, and increased public transport

adoption, while supporting SDG 13 (Climate Action) through optimised operations reducing urban carbon emissions.

5 Research Plan

This research will be conducted over 31 weeks from 9 March 2026 to 14 September 2026 under the course *WQD7025 Data Science Research Project*, which spans Semester II (23 weeks) and Special Semester (8 weeks). The research is structured into five phases, Phase 1 (Weeks 1-6) focuses on data collection and preprocessing of KTM Komuter ridership and meteorological data, Phase 2 (Weeks 7-12) involves feature engineering and baseline model development, Phase 3 (Weeks 13-22) covers STGNN architecture design, implementation and training with ablation studies, Phase 4 (Weeks 23-28) encompasses comprehensive model evaluation and comparative analysis; and Phase 5 (Weeks 29-31) is allocated for thesis writing, final submission and presentation. This timeline is feasible within the master's scope with clear deliverables per phase and sufficient time for iterative refinement.

As for the tools and platforms, the research will be implemented using Python with standard data science libraries (NumPy, Pandas, Scikit-learn) for data preprocessing. Deep learning implementation will utilise PyTorch as the core framework and PyTorch Geometric (PyG) for graph neural network development, while model training will leverage publicly available GPU resources from Google Colab to ensure efficient and accessible computation for STGNN training and experimentation.

References

Farahmand, Zakir H., Konstantinos Gkiotsalitis, and Karst T. Geurs. 2023. 'Predicting Bus Ridership Based on the Weather Conditions Using Deep Learning Algorithms'. *Transportation Research Interdisciplinary Perspectives* 19:100833. doi:10.1016/j.trip.2023.100833.

Jin, Guangyin, Yuxuan Liang, Yuchen Fang, Zezhi Shao, Jincai Huang, Junbo Zhang, and Yu Zheng. 2024. 'Spatio-Temporal Graph Neural Networks for Predictive Learning in Urban Computing: A Survey'. *IEEE Transactions on Knowledge and Data Engineering* 36(10):5388–5408. doi:10.1109/TKDE.2023.3333824.

Li, Ruisen, Liqiang Zhao, Jinjin Tang, Shuixiong Tang, and Zhenxing Hao. 2025. 'Multi-Dimensional Time-Dependent Dynamic Graph Neural Network for Metro Passenger Flow Prediction'. *Applied Intelligence* 55(6):442. doi:10.1007/s10489-025-06346-z.

Liu, Wenzheng, Hongtao Li, Haina Zhang, Jiang Xue, and Shaolong Sun. 2025a. 'Dynamic Spatio-Temporal Graph Fusion Network Modeling for Urban Metro Ridership Prediction'. *Information Fusion* 117:102845. doi:10.1016/j.inffus.2024.102845.

Ma, Zhaobin, Zhiqiang Lv, Zhihao Xu, Rongkun Ye, and Jianbo Li. 2025. 'Spatio-Temporal Traffic Flow Forecasting Based on Second-Order Continuous Graph Neural Network'. *The Computer Journal* 68(9):1300–1314. doi:10.1093/comjnl/bxaf038.

Malaysia Open Data Portal, Government of. n.d. 'Hourly Origin-Destination Ridership: Komuter | Data.Gov.My'. Retrieved 11 January 2026. https://data.gov.my/data-catalogue/ridership_od_komuter.

Malaysian Meteorological Department. n.d. 'myMETdata | Home'. Retrieved 11 January 2026. https://mymetdata.met.gov.my/.

Ren, Peng, Shuang Liu, Beining Qin, Yue Chen, Qi Xu, and Peng He. 2025. 'A Novel Multimodal Deep Learning-Based Direct Ridership Model for Planning-Oriented Demand Prediction in Urban Rail Transit Networks'. *Journal of Transport Geography* 129:104409. doi:10.1016/j.jtrangeo.2025.104409.

Wang, Qingyi, Shenhao Wang, Dingyi Zhuang, Haris Koutsopoulos, and Jinhua Zhao. 2024. 'Uncertainty Quantification of Spatiotemporal Travel Demand With Probabilistic Graph Neural Networks'. *IEEE Transactions on Intelligent Transportation Systems* 25(8):8770–81. doi:10.1109/TITS.2024.3367779.

Wu, Fei, Changjiang Zheng, Muqing Du, Shukang Zheng, Junze Ma, and Ye Lu. 2025. 'LLM4PT: A Large Language Model-Based System for Flexible and Explainable Public Transit Demand Prediction'. *Computers & Industrial Engineering* 209:111449. doi:10.1016/j.cie.2025.111449.

Yazıcıoğlu, Cafer, and Ali Akgüngör. 2025. 'Modeling Short-Term Passenger Flows in Metro and Bus Systems Using Meteorological Data: Deep Learning Model Comparisons'. *Applied Sciences* 15(11):6260. doi:10.3390/app15116260.

Zeb, Muhammad Shah, Muhammad Asif Khan, Muhammad Muzzamil Hussain Khattak, Sameer Ud-Din, Muhammad Faisal Habib, and Muhammad Zaheer Khan. 2025. 'Forecasting Public Transit Ridership amidst COVID-19: A Machine Learning Approach'. *Public Transport* 17(2):391–420. doi:10.1007/s12469-024-00368-5.

Zhang, Shuxin, Jinlei Zhang, Lixing Yang, Feng Chen, Shukai Li, and Ziyou Gao. 2024. 'Physics Guided Deep Learning-Based Model for Short-Term Origin–Destination Demand Prediction in Urban Rail Transit Systems Under Pandemic'. *Engineering* 41:276–96. doi:10.1016/j.eng.2024.04.020.

Zhao, Xinru, Wenhao Yu, and Yifan Zhang. 2025. 'MST-GNN: Graph Neural Network with Multi-Granularity in Space and Time for Traffic Prediction'. *GeoInformatica* 29(3):377–401. doi:10.1007/s10707-024-00532-w.

Zou, Linmu, Zijia Wang, and Rui Guo. 2024. 'Real-Time Prediction of Transit Origin–Destination Flows during Underground Incidents'. *Transportation Research Part C: Emerging Technologies* 163:104622. doi:10.1016/j.trc.2024.104622.